Curriculum Vitae

Caleb K. Harada

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Research Interests *(UAT Keywords)*

Exoplanet detection (489), Habitable planets (695), Exoplanet systems (484), Planet hosting stars (1242), Exoplanet evolution and formation (491, 492), Exomoons (483), Exoplanet rings (494), Exoplanet atmospheres (487)

Education

- **Ph.D. in Astrophysics**, University of California, Berkeley — Expected May 2026
  Dissertation: "*The Astrophysical Environments of Habitable Worlds*"
  Advisor: Prof. Courtney Dressing
- **M.A. in Astrophysics**, University of California, Berkeley — May 2022
- **B.S. in Astronomy w. High Honors** (*Magna Cum Laude*), University of Maryland — May 2020
  Honors Thesis: "*Signatures of Clouds in Hot Jupiter Atmospheres*"
  Advisor: Prof. Eliza M.-R. Kempton
- **B.S. in Physics** (*Magna Cum Laude*), University of Maryland — May 2020

Research Appointments

- **Graduate Student Researcher**, Dept. of Astronomy, UC Berkeley — 2020–Present
- **NSF Graduate Research Fellow**, Dept. of Astronomy, UC Berkeley — 2020–2025
- **Undergraduate Student Researcher**, Dept. of Astronomy, University of Maryland — 2017–2020
- **NSF REU Intern**, Center for Astrophysics | Harvard & Smithsonian — 2019
- **Summer Research Assistant**, Dept. of Astronomy, University of Michigan — 2018
- **Summer Research Assistant**, Dept. of Astronomy & Astrophysics, University of Chicago — 2017

Awards, Scholarships & Certificates

- AAS Rodger Doxsey Dissertation Travel Prize (*$400*) — 2026
- UC Berkeley Certificate in Teaching & Learning in Higher Education — 2025
- UC Berkeley Outstanding Graduate Student Instructor Award (*$1k*) — 2022
- NSF Graduate Research Fellowship (*$34k–$37k annually for 3 years*) — 2020
- UMD University Honors Certificate — 2019
- NSF Research Experience for Undergraduates (*$5k*) — 2019
- Maryland Space Grant Scholarship (*$2k*) — 2019
- UMD Monroe Martin Undergraduate Research Showcase Award (*$250*) — 2018, 2019
- UMD Angelo Bardasis Memorial Scholarship (*$2.5k*) — 2018, 2019
- UMD President's Scholarship (*$5k annually for 4 years*) — 2016

PEER-REVIEWED PUBLICATIONS *(21 total; 4 first author)* Search SciX (ADS)

Significant Contributions

21. **Harada**, Dressing, Turtelboom, et al., "*SPORES-HWO. II. Companion Mass Limits and Updated Planet Properties for 120 Future Exoplanet Imaging Targets from 35 yr of Precise Doppler Monitoring*," 2025, AJ, 170, 343
20. **Harada**, Dressing, Kane, & Ardestani, "*Setting the Stage for the Search for Life with the Habitable Worlds Observatory: Properties of 164 Promising Planet Survey Targets*," 2024, ApJS, 272, 30
19. **Harada**, Dressing, Alam, et al., "*Stability and Detectability of Exomoons Orbiting HIP 41378 f, a Temperate Jovian Planet with an Anomalously Low Apparent Density*," 2023, AJ, 166, 208
18. **Harada**, M.-R. Kempton, Rauscher, et al., "*Signatures of Clouds in Hot Jupiter Atmospheres: Modeled High-Resolution Emission Spectra from 3D General Circulation Models*," 2021, ApJ, 909, 85
17. Tuchow, **Harada**, Mamajek, et al., "*HWO Target Stars and Systems: A Prioritized List of Potential Stellar Targets for the Habitable Worlds Observatory's ExoEarth Survey*," 2025, PASP, 137, 104402
16. Desai, Turtelboom, **Harada**, et al., "*The TESS-Keck Survey. XVIII. A Sub-Neptune and Spurious Long-period Signal in the TOI-1751 System*," 2024, AJ, 167, 194
15. Duck, **Harada**, Harrell, Morris, Williams, Crossfield, Werner, & Deming, "*K2, Spitzer, and TESS Transits of Four Sub-Neptune Exoplanets*," 2021, AJ, 162, 136

Co-authored Publications

14. Kane, Bott, Gordon, et al. (inc. **Harada**), "*Imaging Venus-like Worlds: Spectral, Polarimetric, and UV Diagnostics for the Habitable Worlds Observatory*," 2026, PASP, 138, 024404
13. Li, Kane, Blunt, & **Harada**, "*Radial Velocity Strategies for the Orbital Refinement of Exoplanet Direct Imaging Targets*," 2025, PASP, 137, 104401
12. Turtelboom, Dietrich, Dressing, & **Harada**, "*Searching for Additional Planets in TESS Multiplanet Systems: Testing Empirical Models Based on Kepler Data*," 2025, AJ, 170, 3
11. Mayo, Fortenbach, Louie, Dressing, Turtelboom, Giacalone, & **Harada**, "*Detection of H_2O and CO_2 in the Atmosphere of the Hot Super-Neptune WASP-166b with JWST*," 2025, AJ, 170, 50
10. Wittenmyer, Errico, Holt, Horner, **Harada**, Kane, Li, & Fetherolf, "*Optimising Radial Velocity Detection Limits for Southern Habitable Worlds Observatory Targets*," 2025, MNRAS, 539, 1
9. Grouffal, Santerne, Bourrier, et al. (inc. **Harada**), "*The Star HIP 41378 Potentially Misaligned With Its Cohort of Long-period Planets*," 2025, A&A, 701, A173
8. Blunt, Nielsen, Newton, et al. (inc. **Harada**), "*A Statistical Method for Constraining the Capability of the Habitable Worlds Observatory to Understand Ozone Onset Time in Earth Analogs*," 2025, JATIS, 11, 4
7. Sagynbayeva, Abbas, Kane, et al. (inc. **Harada**), "*Requirements for Joint Orbital Characterization of Cold Giants and Habitable Worlds with Habitable Worlds Observatory*," 2025, AJ, 170, 208
6. Kane, Li, Turnbull, Dressing, & **Harada**, "*Dynamical Viability Assessment for Habitable Worlds Observatory Targets*," 2024, AJ, 168, 195
5. Turtelboom, Weiss, Dressing, et al. (inc. **Harada**), "*The TESS-Keck Survey. XI. Mass Measurements for Four Transiting Sub-Neptunes Orbiting K-dwarf TOI 1246*," 2022, AJ, 163, 293
4. Malsky, Rauscher, M.-R. Kempton, Roman, Long, & **Harada**, "*Modeling the High-Resolution Emission Spectra of Clear and Cloudy Nontransiting Hot Jupiters*," 2021, ApJ, 923, 62
3. Roman, M.-R. Kempton, Rauscher, **Harada**, Bean, & Stevenson, "*Clouds in Three-Dimensional Models of Hot Jupiters Over a Wide Range of Temperatures I: Thermal Structures and Broadband Phase Curve Predictions*," 2021, ApJ, 908, 101

2. Vissapragada, Knutson, Jovanovic, **Harada**, et al., "*Constraints on Metastable Helium in the Atmospheres of WASP-69b and WASP-52b with Ultra-Narrowband Photometry*," 2020, AJ, 159, 278
1. Collins, Collins, Pepper, et al. (inc. **Harada**), "*The KELT Follow-Up Network and Transit False Positive Catalog: Pre-vetted False Positives for TESS*," 2018, AJ, 156, 234

In-Prep or Non-Refereed Manuscripts

3. **Harada**, Dressing, Kane, et al., "*Discovery of an Eccentric Giant Companion in the Habitable Zone of a Nearby Sun-like Star*," in prep
2. Dressing, Savel, Giacalone, et al. (inc. **Harada**), "*Cleaning up the TOIs I: ShARCS Follow-up Adaptive Optics Imaging of 603 TESS Targets*," Submitted to AJ
1. Li, Meyer, D'Angiolillo, et al. (inc. **Harada**), "*A Century of Radial Velocity and Astrometric Monitoring of 70 Oph AB: New PFS Data and Constraints on Planetary Companions*," Submitted to AJ

Research Presentations

Contributed talk	**AAS Meeting #247**, Phoenix, AZ	Jan. 5, 2026
Invited talk	**UC Berkeley CIPS Seminar**, Berkeley, CA	Dec. 3, 2025
Invited talk	**UC Riverside Astrobiology Seminar**, Riverside, CA	Nov. 14, 2025
Invited talk	**Caltech/IPAC Lunch Seminar**, Pasadena, CA	Nov. 12, 2025
Invited talk	**Harvard/CfA Special Seminar**, Cambridge, MA	Oct. 16, 2025
Invited talk	**UC Santa Cruz PLUNCH Seminar**, Santa Cruz, CA	Sep. 29, 2025
Contributed talk	**HWO25 Conference**, Washington, DC	Jul. 30, 2025
Poster	**Sagan Summer Workshop 25**, Pasadena, CA	Jul. 21-25, 2025
Contributed talk	**Bay Area Exoplanet Meeting #48**, Santa Cruz, CA	Jul. 18, 2025
Invited talk	**HWO TSS-SWG Multiplicity Group Meeting**, Remote	Jul. 16, 2025
Poster	**EPRV6 Conference**, Porto, Portugal	Jul. 1-3, 2025
Poster	**Know Thy Star; Know Thy Planet 2**, Pasadena, CA	Feb. 3-7, 2025
Contributed talk	**AAS Meeting #245**, National Harbor, MD	Jan. 15, 2025
Invited talk	**Yale Exoplanets & Stars Seminar**, New Haven, CT	Oct. 30, 2024
Invited talk	**MIT Kavli Monday Afternoon Talk**, Cambridge, MA	Oct. 28, 2024
Poster	**Sagan Summer Workshop 24**, Pasadena, CA	Jul. 22-26, 2024
Poster	**Exoplanets 5 Conference**, Leiden, The Netherlands	Jun. 20, 2024
Poster	**Extreme Solar Systems V**, Ōtautahi, Aotearoa	Mar. 17-21, 2024
Contributed talk	**AAS Meeting #243**, New Orleans, LA	Jan. 10, 2024
Invited talk	**AAS #243 HWO Splinter Session**, New Orleans, LA	Jan. 10, 2024
Invited talk	**UC Santa Cruz PLUNCH Seminar**, Santa Cruz, CA	Oct. 31, 2022
Contributed talk	**UC Berkeley Astronomy Lunch Talk**, Berkeley, CA	Oct. 20, 2022
Contributed talk	**Bay Area Exoplanet Meeting #36**, Remote (COVID-19)	Mar. 5, 2021
Poster	**AAS Meeting #235**, Honolulu, HI	Jan. 6, 2020
Contributed talk	**CfA/SAO Summer REU Symposium**, Cambridge, MA	Aug. 8, 2019
Poster	**Chesapeake Bay Exoplanet Meeting #4**, College Park, MD	Feb. 15, 2019
Poster	**AAS Meeting #233**, Seattle, WA	Jan. 8, 2019

Approved Observing Proposals *(** = Science PI)*

****2.4-m Automated Planet Finder (APF)**, Lick Observatory, "*Probing Planetary System Architectures Around Neglected Nearby Stars in Preparation for HWO*," (Levy; 9 nights across 2025A/B, 2026A; PI: **Harada**)

****10-m Keck I Telescope**, Mauna Kea Observatory, "*Confirming an Eccentric Sub-Saturn Planet Around the Bright Nearby Star HD 219623*," (KPF; 0.25 nights across 2025A; PI: **Harada**)

10-m Keck I Telescope, Mauna Kea Observatory, "*Piecing Together the Puzzling TOI-880 System*," (KPF; 1.0 nights across 2025B; PI: Turtelboom)

2.4-m Automated Planet Finder (APF), Lick Observatory, "*The APF Survey of the Habitable Worlds Observatory's Neglected Targets*," (Levy; 8 nights across 2025A/B; PI: Holden)

3-m Shane Telescope, Lick Observatory, "*Looking for Close Stellar Companions to Potential Targets for Future Searches for Life with HWO*," (ShARCS AO, 10 nights across 2025A/B; PI: Dressing)

8.1-m Gemini-N Telescope, Mauna Kea Observatory, "*Characterizing TOI 2104, one of the highest-multiplicity TESS systems*," (MAROON-X, 40 hrs across 2024B, 2025B; PI: Turtelboom)

3.5-m WIYN Telescope, Kitt Peak National Observatory, "*Characterizing TOI 2104, one of the highest-multiplicity TESS systems*," (NEID, 18 hrs across 2024B; PI: Turtelboom)

10-m Keck I Telescope, Mauna Kea Observatory, "*The Obliquity of the HIP 41378 Planetary System*," (HIRES; 18 hrs in 2022B; PI: Dressing)

1-m Nickel Telescope, Lick Observatory, "*Validating Netpune-size planets around A-type stars*," (Direct Imaging Camera; 43 nights across 2021B, 2022A/B; PI: Giacalone)

James Webb Space Telescope (JWST), "*Transmission Spectroscopy of the Super-Neptune WASP-166b*," (NIRISS & NIRSpec; 23 hrs Cycle 1 GO Program 2062; PI: Mayo)

3-m Shane Telescope, Lick Observatory, "*A Closer Look at the Host Stars of Transiting Planets*," (ShARCS AO, 25 nights across 2021A/B, 2022A; PI: Dressing)

Professional Service

- **Scientific Reviewer**, The Astronomical Journal (AJ); Astronomy & Astrophysics (A&A) — 2025–Present
- **Chambliss Poster Competition Judge**, AAS Winter Meetings — 2024, 2025, 2026
- **Catalogs & Databases Co-Lead**, NASA HWO Target Stars & Systems Group — 2024–2025
- **Target Stars & Systems Laiason**, NASA HWO Exoplanet Demographics Group — 2024–2025

Departmental Service

- **Co-Organizer**, UC Berkeley Astro-Q — 2023–Present
- **Grad Representative**, UC Berkeley Astrophysics Graduate Admissions Committee — 2025, 2026
- **Lead Organizer**, UC Berkeley Astronomy RV Hunters Community Science Project — 2025
- **Grad Representative**, UC Berkeley Astronomy DEI & Climate Committee — 2021–2022
- **Grad Representative**, UC Berkeley Astro Grad Prospective Visit Planning Committee — 2021

Teaching Experience

- **The Planets /** ASTRON-C12 (*UC Berkeley undergrad non-majors*)
  - Guest Lecturer — Apr. 1, 2025
  - Head Graduate Student Instructor — Spring 2023
- **Introduction to Astrophysics /** ASTRON-7A (*UC Berkeley undergrad majors*)
  - Guest Lecturer — Nov. 14, 2024, Sep. 5, 2023, Oct. 18, 2022
  - Graduate Student Instructor — Fall 2021
- **Atmospheric Physics and Dynamics /** EPS-C181 (*UC Berkeley undergrad majors*)
  - Course Reader — Fall 2022
- **Introductory Astrophysics II /** Astro-121 (*University of Maryland undergrad majors*)
  - Lab Teaching Assistant (remote due to COVID-19) — Spring 2020

Course Tutor — Spring 2018

- Observational Astronomy / Astro-310 (*University of Maryland undergrad majors*)
Lab Teaching Assistant — Fall 2018, 2019

Mentoring Programs

- Astro Grad Mentoring (*for UC Berkeley 1st-year grads*)
Senior Graduate Student Mentor — 2025–Present

- Mathematical & Physical Sciences (MPS) Scholars (*for UC Berkeley undergrads*)
Graduate Student Mentor — 2024–Present
Workshop Panelist — Dec. 3, 2024, Oct. 10, 2024, Aug. 8, 2024

- Cal-Bridge (*California-wide bridge program for underrepresented undergrads*)
Graduate Student Tutor — Fall 2024

- POWER Bay Area (*mentoring program for underrepresented community college students*)
Mentor Coordinator — 2022–2023

- Astronomy Peer Mentoring (APM) (*for UMD undergrads*)
Undergraduate Student Mentor — 2018–2020

Public Outreach

- Letters to a Pre-Scientist (*K–12 outreach program for low-income students*)
Pen Pal (letter writer) — 2024–Present

- The Compass Project (*UC Berkeley talk series for undergraduates in the physical sciences*)
Guest Lecturer — Oct. 25, 2024

- Scientist in Every Florida School (*K–12 outreach program for Florida public schools*)
Guest Speaker — Oct. 7, 2024

- University of Maryland Observatory (*bimonthly public open house events and observing*)
Observatory Staff Member — 2017–2020
Observatory Volunteer — 2016–2017

- Maryland Day (*campus-wide UMD public outreach event*)
Astronomy Department Volunteer — 2017–2019

Students Advised

- Christopher Chin, UC Berkeley undergrad (began 2025)
- Aidan Fernandez, USC & Mt. San Antonio Community College undergrad (began 2025)
- Shivani Kulkarni, UC Berkeley undergrad (began 2025)
- Emerald Lin, UC Berkeley undergrad (began 2025)
- Nykole Liu, UC Berkeley undergrad (began 2025)
- Remy Liu, UC Berkeley undergrad (began 2025)
- Abhi Nathan, UC Berkeley undergrad (began 2025)
- Bahareh A. Ardestani, Sonoma State University undergrad (began 2023; co-advised)
- Ryan Hwangbo, UC Berkeley undergrad (began 2023; co-advised)
- Anmol Desai, UC Berkeley undergrad (began 2022; co-advised)

References available upon request.

Updated March 2026.